

SATYA VENKATA MANIKANTA M.

Software Development Engineer – Java | AWS | Generative AI | Distributed Systems
Texas, USA | 2816685626 | satyamanikantam1@gmail.com | LinkedIn | Portfolio

PROFESSIONAL SUMMARY:

- **Java Full Stack Engineer with 4+ years of experience** designing, developing, and scaling enterprise-grade microservices and cloud-native applications using **Java 17, Spring Boot 3.x, React/Angular, and AWS** across finance, AI, and analytics domains.
- **Proven expertise** in building **high-performance, event-driven architecture** and **real-time data pipelines** with **Kafka Streams, REST, and gRPC**, processing over **1M+ daily transactions** with **99.98% uptime**.
- **Hands-on experience integrating Generative AI (LLMs – GPT, BERT, LangChain4j)** into Java microservices for **personalized recommendations, analytics automation, and adaptive user experiences**, bridging AI with enterprise-grade systems.
- **Skilled in designing intuitive front-end dashboards** using **React and Angular**, transforming complex data into actionable, real-time insights with modern UI/UX principles.
- **Strong foundation in system design, concurrency, data structures, and algorithmic optimization**, ensuring scalability, reliability, and maintainability in distributed environments.
- **Experienced across AWS and Azure ecosystems**, including **ECS, EKS, Lambda, AKS, and DynamoDB**, leveraging **Terraform, Helm, and GitHub Actions** for automated, zero-downtime deployments and 40% faster delivery cycles.
- **Security-first and performance-driven engineer** implementing **OAuth 2.1, JWT, Keycloak, TLS/SSL**, and API security policies to deliver resilient, compliant systems on a scale.
- **Innovative, collaborative, and business-focused technologists** passionate about blending **engineering excellence with AI innovation**, driving continuous improvement, operational efficiency, and measurable business outcomes.

TECHNICAL SKILLS:

Programming Languages:	Java (primary), Python, JavaScript, TypeScript, SQL, Shell
Frameworks & Libraries:	Spring Boot, Spring Cloud, Hibernate, JPA, Microservices Architecture
APIs & Web Services:	REST APIs, GraphQL, gRPC, Swagger, JSON, XML, Postman, Spring Cloud Gateway
Messaging & Streaming:	Apache Kafka, Kafka Streams, Avro, AWS Kinesis, Spring Cloud Stream
AI & Machine Learning:	Generative AI (LLMs – GPT, BERT), PyTorch, TensorFlow, Spark ML, Airflow Pipelines
Authentication & Security:	OAuth 2.1, JWT, Spring Security, Keycloak, TLS/SSL, SonarQube, Checkmarx
Database & Caching:	PostgreSQL, Oracle, MongoDB, Redis, Hazelcast, ElastiCache
DevOps & CI/CD:	Jenkins, GitHub Actions, GitLab CI/CD, Docker, Terraform, Helm, Maven
Containerization & Orchestration:	Docker, Kubernetes, OpenShift, AKS, ECS, EKS
Cloud Platforms:	AWS (Lambda, S3, Aurora Postgres, DynamoDB, CloudWatch), Microsoft Azure
Testing & Automation:	JUnit, TestNG, Mockito, Selenium, REST Assured, Cucumber (TDD/BDD), Code Reviews
Observability & Monitoring:	Prometheus, Grafana, ELK, Dynatrace, Jaeger
Tools & Collaboration:	Git, GitHub, Jira, Confluence, IntelliJ, VS Code
Operating Systems & Scripting:	Linux/Unix, Bash/Shell scripting, data analysis & performance tuning

PROFESSIONAL EXPERIENCE:

Charles Schwab – TX | May 2024 – Present

Software Engineer (Software Development Engineer – Full Stack)

Roles and Responsibilities:

- **Architected distributed, event-driven microservices** for the **Real-Time TradeOps Platform**, enabling **risk-aware order management** and instant trade validation using **Java 17 and Spring Boot 3.x**, deployed across **AWS ECS and Azure AKS** with **PostgreSQL, Oracle, and Aurora Postgres** for multi-region reliability and fault tolerance.
- **Engineered high-throughput Kafka Streams pipelines** that processed **millions of trade events per day**, achieving **sub-second latency** for risk evaluation, settlement updates, and market data ingestion.
- **Delivered and scaled REST, GraphQL, and gRPC APIs** secured with **OAuth 2.1, JWT, and Spring Security**, ensuring **compliant cross-domain communication** between trading engines, risk services, and reporting modules.
- **Built fully automated CI/CD and IaC pipelines** using **Terraform, AWS CDK, Helm, Jenkins, and GitHub Actions**, enabling **zero-downtime deployments** and **40 % faster release cycles** across dev, staging, and production.
- **Integrated Generative AI and LLM modules** (GPT, BERT via PyTorch/TensorFlow) to augment **trade surveillance, risk prediction, and pattern recognition**, improving anomaly detection accuracy and decision support.
- **Optimized data-layer performance** through **Redis, Hazelcast, and AWS ElastiCache**, cutting query latency by **60 %** and boosting throughput for high-volume trade lookups and real-time dashboards.
- **Implemented observability and SRE practices** with **Prometheus, Grafana, ELK, Dynatrace, and Jaeger**, maintaining **99.98 % uptime** and enabling proactive alerting for trade-critical microservices.
- **Mentored junior engineers** and led **architecture and code reviews**, raising unit-test coverage to **> 92 %** while enforcing clean-code, SOLID, and scalability standards across backend teams.
- **Applied fault-tolerant design patterns** (Circuit Breaker, Retry, Bulkhead) to isolate failures and maintain consistency during peak trading hours, improving **system resilience and operational continuity**.

HCLTech – India | June 2020 - July 2023

Software Engineer (Full Stack Developer)

Roles and Responsibilities:

- **Engineered and modernized legacy healthcare systems** into **modular, secure microservices** using **Java 17, Spring Boot, and PostgreSQL**, enabling high-volume claims processing and seamless data exchange between providers and payers.
- **Implemented robust authentication and authorization** using **Spring Security, OAuth 2.0, and JWT**, ensuring **HIPAA and PCI-DSS compliance** across all service integrations.
- **Developed event-driven transaction pipelines** with **Apache Kafka and Redis**, synchronizing real-time HL7/FHIR data flows and inter-system communication between EHR and claims modules.
- **Optimized backend performance** through **SQL tuning, Redis caching, and Elasticsearch indexing**, reducing query latency by 25 % and improving API response times under peak load.
- **Automated CI/CD pipelines** via **Jenkins, GitLab CI/CD, Docker, and OpenShift**, accelerating deployments and increasing release frequency for multi-environment delivery.
- **Containerized and deployed microservices** on **AWS ECS and Kubernetes**, enhancing uptime, scalability, and fault tolerance for critical healthcare transactions.
- **Designed and executed comprehensive integration and API tests** using **Postman, TestNG, and REST Assured**, ensuring system stability and data integrity across high-volume workloads.
- **Collaborated with architecture, QA, and DevOps teams** in **Agile/Scrum** sprints to enforce design standards, code reviews, and continuous delivery best practices for enterprise clients.

EDUCATION:

Master of Computer Science - University of North Texas, Denton, TX, USA

Bachelor of Engineering - Hindustan Institute of Technology, Coimbatore, India

PORTFOLIO & ONLINE PRESENCE:

Portfolio: manikantamedidi.com – Showcasing real-time trading and healthcare projects built with Java, Spring Boot, React, Kafka, and Generative AI integrations.